

Topic: Types of Queue Circular Queue, Priority Queue

Q. Explain the limitations of a simple linear queue. How do a Circular Queue and a Priority Queue overcome specific challenges? Describe their working principles.

> **Definition to Remember**

> A **Priority Queue** is a data structure that stores tasks to be executed in the future. The tasks are stored in the queue based on their priority.

Queue Types: Circular & Priority Queues

In a standard linear array queue, once the `rear` pointer reaches the maximum index ($\text{SIZE} - 1$), no new elements can be enqueued. This happens **even if spaces are empty at the front** due to earlier dequeue operations, leading to severe **memory wastage**.

2. Circular Queue

Solution: The queue forms a logical circle. The next position after the last index is index 0.

- **Working Principle:** Uses the **modulo operator** ($\% \text{SIZE}$) to wrap pointers around.

- Next

[0] <--

3. Priority Queue

Problem: A standard queue strictly processes the oldest element first. Some real-world tasks (like CPU interrupts) are urgent and cannot wait in a FIFO line.

Solution:

- **Working Principle:**

- Element

> **Must-Write Points (for 10 marks)**

> 1. **Priority Queue**

> **Quick Recall**

> `Line
by high

Complete Executable C Code: Circular Queue Array Implementation

```
```c
```

#includ

```
#define MAX 5
```

```
struct CircularQueue {
```

int item

```
void initQueue(struct CircularQueue *q) {
```

q->from

```
int isFull(struct CircularQueue *q) {
```

return

```
int isEmpty(struct CircularQueue *q) {
```

return

```
void enqueue(struct CircularQueue *q, int val) {
```

if (isFu

```
int dequeue(struct CircularQueue *q) {
```

```
if (isEn
```

```
void display(struct CircularQueue *q) {
```

```
if (isEn
```

```
int main() {
```

```
struct
```

```
enqueue(&q, 10);
```

```
enque
```

```
printf("Dequeued: %d\n", dequeue(&q));
```

```
printf("
```

```
enqueue(&q, 50);
```

```
enque
```

```
return 0;
```

```
}
```